

Final Project Report

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Project Overview:

The overall purpose of this project was to use a tool to design, configure, and test a small business computer network using my chosen tool, Cisco Packet Tracer. The network is simulating a real office environment where there are multiple client computers communicating with a central server through a router and switch. The project shows how common networking protocols work together to provide network connectivity and communication.

The network was configured to implement core networking concepts, like DHCP, DNS, ARP, HTTP, ICMP, TCP, and IP addressing. Testing and pinging was performed to verify that all devices could successfully communicate and that the implemented services functioned correctly.

Network Design:

The network was designed to represent a small business office containing one router, one switch, one server, and three client computers. The router serves as the default gateway for the local network, while the switch provides connectivity between all devices. The server hosts several network services including DHCP, DNS, and HTTP. The client computers obtain network connectivity through the switch and communicate with the server to access network resources.

The topology consists of:

- Cisco 2911 Router
- Cisco 2960 Switch
- Three client PCs
- Only 1 Server

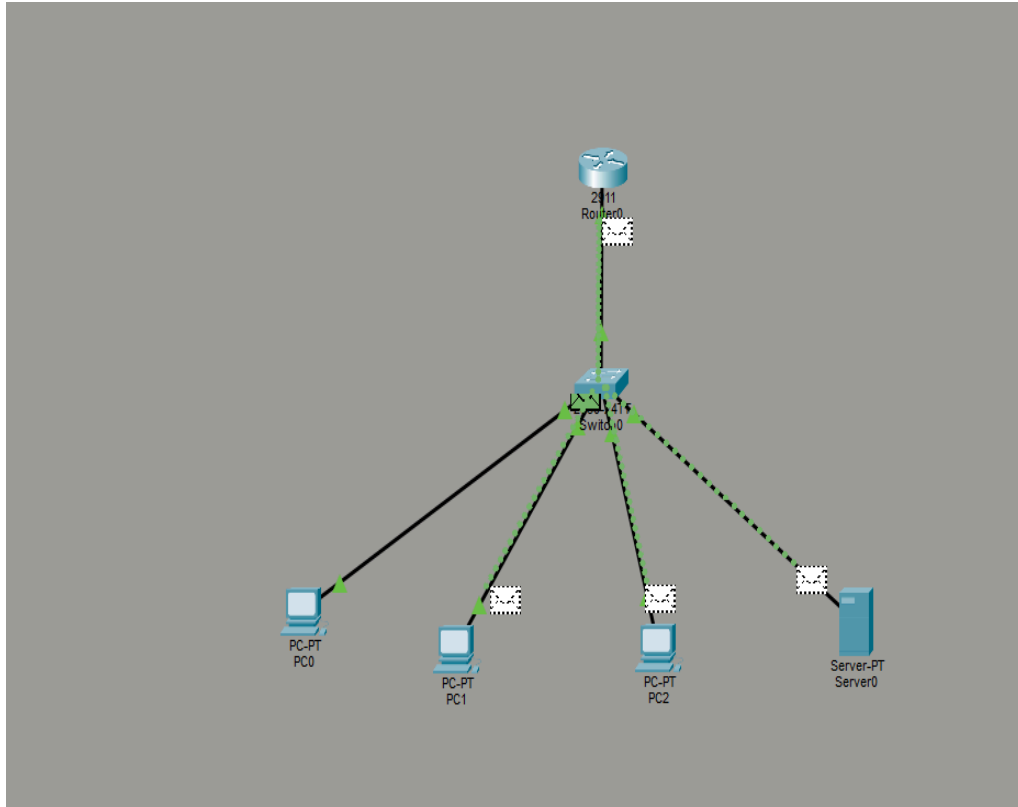


Figure 1:Network Topology

IP Addressing:

My network uses the private IPv4 address space 192.168.1.0/24.

Router0: 192.168.1.1

Server0: 192.168.1.10

PC0: DHCP Assigned (192.168.1.2)

PC1: DHCP Assigned

PC2: DHCP Assigned

The router was configured as the main default gateway, it was allowing client devices to communicate across the local network. The server was assigned a static IP address so that network services would always be available at a consistent address

DHCP Configuration:

Dynamic Host Configuration Protocol (DHCP) was configured on the server to automatically assign IPv4 addresses to client computers. A DHCP address pool named as OfficePool was created with the

appropriate network information, including the subnet mask, DNS server, and default gateway. The DHCP service successfully assigned an IP address to client devices during testing.

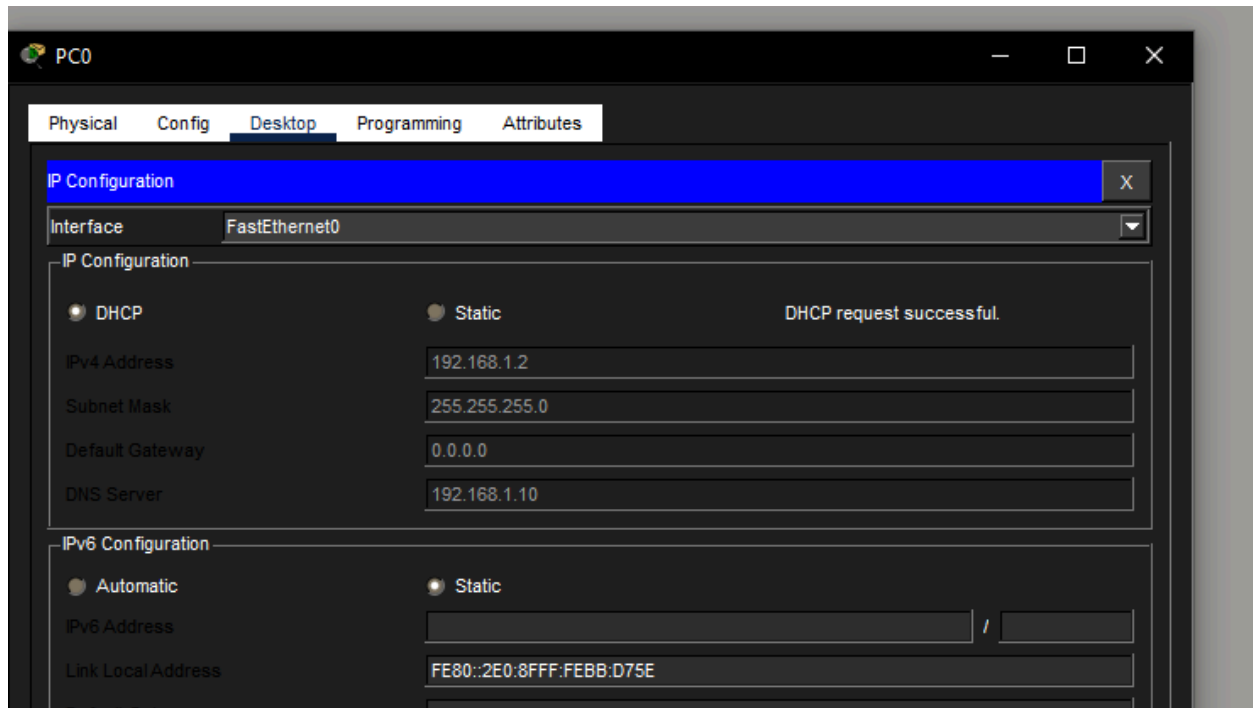


Figure 2: DHCP Configuration

DNS Configuration:

Domain Name System (DNS) was configured on the server to provide name resolution. An A Record was created mapping the hostname: www.company.local to the server's IP address: 192.168.1.10. This allowed users to access the web server using a domain name instead of typing the IP address directly.

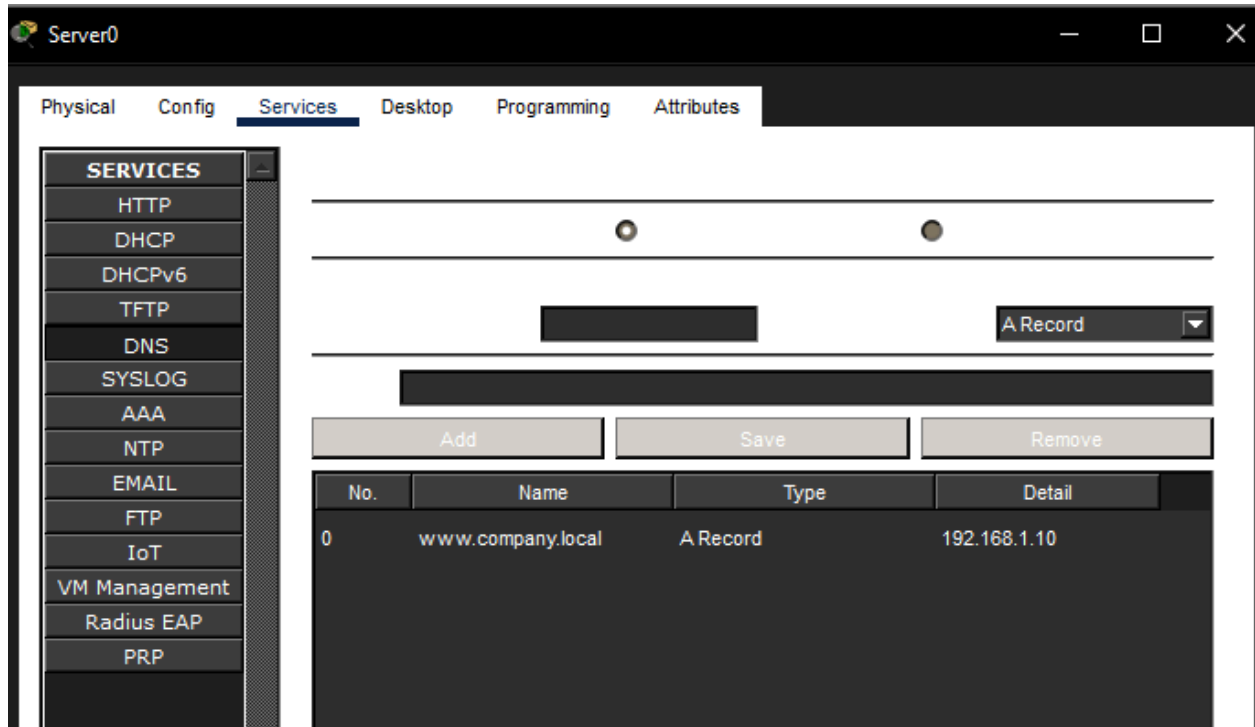


Figure 3: DNS Configuration

HTTP Webserver:

The server was configured to host a simple web page using the built in HTTP service in packet tracer. The client computers successfully accessed the website using the URL: <http://www.company.local> The successful loading of my webpage showed good interaction between the DNS and HTTP services.

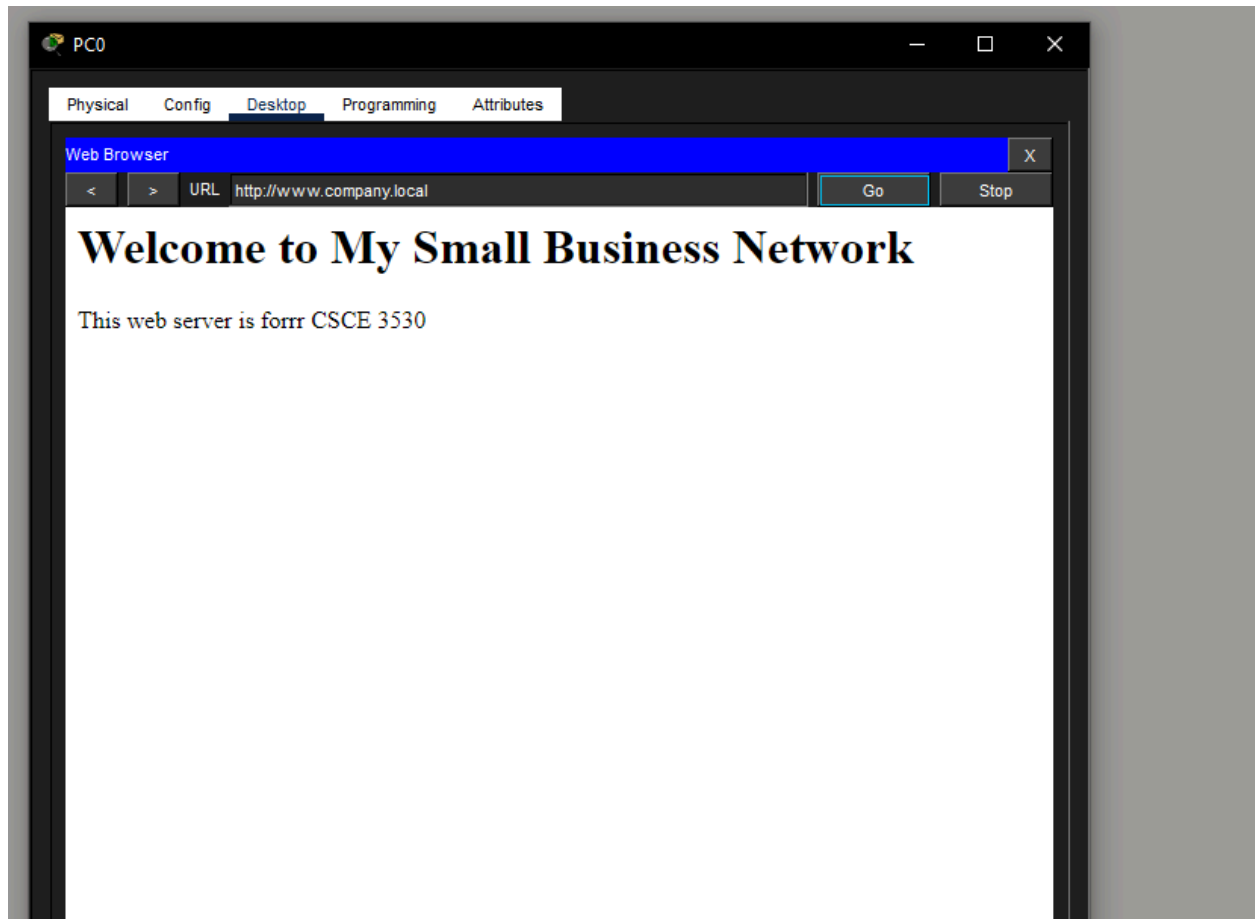


Figure 4: Web Server

Network Testing:

Several tests were performed to verify network functionality. The ping command was used to verify communication between client devices, the router, and the server. All devices successfully responded to ICMP Echo Requests with zero packet loss.

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 5: Successful Pings

Simulation Analysis:

Packet Tracer Simulation Mode was used to observe packets traveling throughout the network. During the simulation, packet movement between the client computers, switch, router, and server was observed. HTTP communication between the client and server was successfully captured. The simulation demonstrated that devices were communicating correctly using the configured network services.

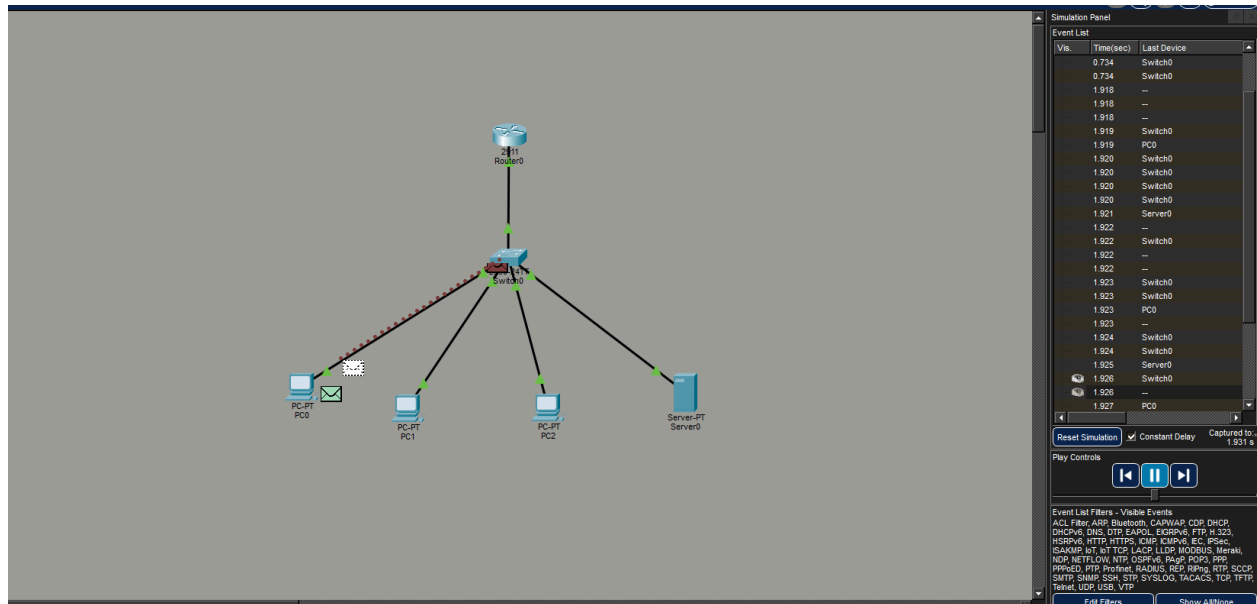


Figure 6: Simulation Mode

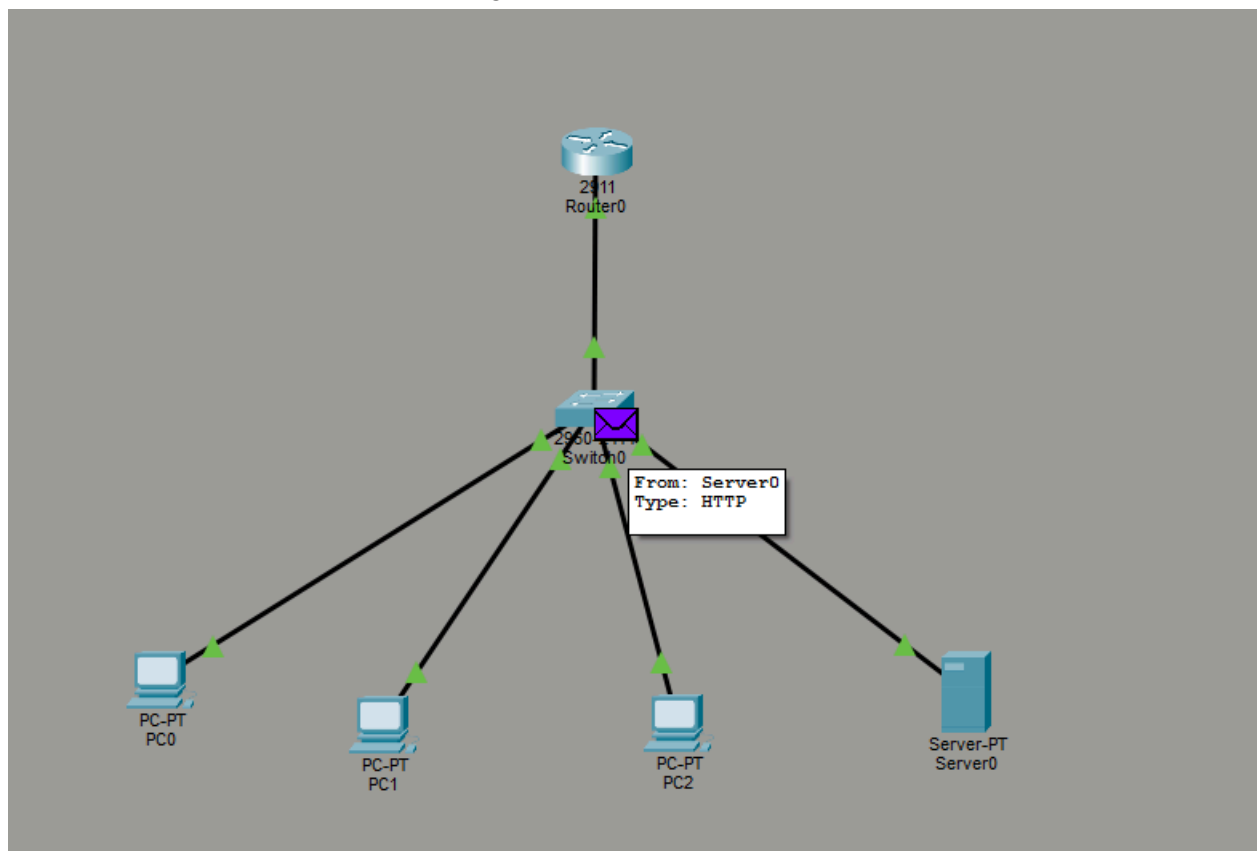


Figure 7: HTTP Packet

Protocols Demonstrated:

In the project I have successfully demonstrated the following networking protocols:

- DHCP: Automatic IP address assignment
- DNS: Hostname resolution
- ARP: MAC address discovery during communication
- HTTP: Client access to the web server
- ICMP: Connectivity testing using ping
- TCP: Reliable communication during HTTP sessions
- IPv4 Addressing: Static and dynamic address configuration

Challenges Encountered:

I had very many configuration issues that were encountered during development of the network since I was new to the tool. Duplicate IP addresses were causing address conflicts between devices. This was resolved by me assigning unique addresses to each device.

Conclusion:

Overall, the project demonstrated the design and implementation of a small business computer network using tool Cisco Packet Tracer. The completed network provided communication between client computers and a server while implementing networking services like DHCP, DNS, HTTP, ICMP, and TCP. Testing confirmed successful communication between all devices, automatic IP address assignment, hostname resolution, and web server accessibility. The project gave practical experience configuring and troubleshooting a functional client server network.